Multiscale and Multidimensional Weighted Network for Salient Object Detection in Optical Remote Sensing Images

Lamei Di, Bin Zhang, *Member, IEEE*, and Yiming Wang

***Abstract*— Recent advancements have significantly benefited in the area of salient object detection in optical remote sensing images (ORSI-SOD). Given the varying spatial resolutions and complex scenes characteristic of optical remote sensing images (ORSIs), leveraging and integrating features across scales is vital. However, excessive feature integration can introduce significant noise and result in inaccurate saliency mapping. To address this issue, we propose the multiscale and multidimensional weighted network for ORSI-SOD (WeightNet). The network adopts a two-stage design where the first stage generates multiscale weighted information, and the second stage conducts indirect multiscale feature weighted fusion. This design skillfully avoids the lack of scale adaptation and noise interference that may arise from direct multiscale feature fusion. Furthermore, to enhance feature fusion and target localization precision, we introduce the multiscale weighted feature aggregation module (MWFAM) and the multidimensional feature guidance module (MDFGM). MWFAM facilitates multiscale feature fusion while minimizing noise from cross-layer interactions. MDFGM specializes in precise target localization and enhancement of detail and edge information. Additionally, the introduction of a multiscale parallel decoder (MPD) significantly boosts the decoder's capability in identifying targets across various scales. Extensive qualitative and quantitative evaluations on three public ORSI datasets demonstrate the effectiveness and superiority of WeightNet over contemporary state-of-the-art models.**

***Index Terms*— Feature weighted, multiscale, optical remote sensing images (ORSIs), salient object detection (SOD).**

I. Introduction

THE task of salient object detection (SOD) is to detect the objects/regions in an image or video that are of most interest to the human eyes based on visual salient features [1], [2]. The goal of SOD is to identify salient objects/regions and to generate a pixel-level salient map (SM), which is essentially a segmentation task [3], [4]. Due to its scalability, SOD has been widely used in computer vision domains, such as image/video segmentation [5], [6], image/video compression [7], [8], visual tracking [9], image editing [10], image quality assessment [11], target classification [12], and so on. Successful applications in these vision tasks have brought SOD to the attention of the world.

SOD for natural scene images (NSI-SOD) has made satisfactory development in the past decades, especially in the years of rapid development of deep learning techniques [13], [14], [15]. However, NSI and ORSI differ in terms of resolution, object type, object scale, and target scene. ORSI is characterized by a wide range of coverage, diverse textures and visual appearances, different object scales and orientations, and overwhelming background context redundancy [16], [17], which leads to many difficulties in the study of SOD for optical remote sensing imagery (ORSI-SOD) still. Researchers consider applying the excellent NSI-SOD solutions directly to ORSI-SOD, and the results are often unsatisfactory. However, with the increasing interest in ORSI as it plays an important role in military, astronomical, and rescue applications, there is an urgent need for a solution dedicated to ORSI-SOD. Existing ORSI-SOD methods can be categorized into traditional and CNN-based methods. Traditional methods contain a large number of specific artificial features obtained based on classical principles and perform poorly in complex ORSI scenarios. CNN-based ORSI-SOD methods are able to learn features better and significantly improve the detection accuracy of salient objects. And, the work in this article is a study based on CNN-based methods.

Currently, CNN-based SOD methods mainly focus on finding more effective strategies for multiscale feature interaction [18]. In some of the existing approaches, features of corresponding layers in the encoder are connected to the decoder through the transport layer to realize multiscale feature interaction [see Fig. 1(a) and (e)] [19], [20]. However, since single-scale features can only describe scale-specific information, the continuous accumulation of deeper features leads to a weakening of the ability to represent details in shallow features. In order to utilize features at multiple scales, some approaches employ a fully connected approach to aggregate multilayer features [see Fig. 1(b) and (f)] [21], [22]. However, this strategy suffers from insufficient resolution balance, leading to inadequate feature interaction. In order to address the interaction balance of multiscale features, some methods use neighboring feature fusion [see Fig. 1(c)] [23], [24], but still suffer from insufficient feature interaction coverage. The structure in Fig. 1(d) proposes a method to integrate all scale features, but too much feature integration may lead to problems such as high computational cost, high noise, and

Manuscript received 6 November 2023; revised 7 March 2024 and 26 April 2024; accepted 11 May 2024. Date of publication 20 May 2024; date of current version 31 May 2024. This work was supported in part by the Key Research and Development Program of Shaanxi under Grant 2022GY-075 and in part by the National Key Projects of China under Grant 2021XJTU0040. *(Corresponding author: Bin Zhang.)*

The authors are with the School of Software Engineering, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: dilamei@stu.xjtu.edu.cn; bzhang82@xjtu.edu.cn; yimingwang@stu.xjtu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3403268

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:13:46 UTC from IEEE Xplore. Restrictions apply.

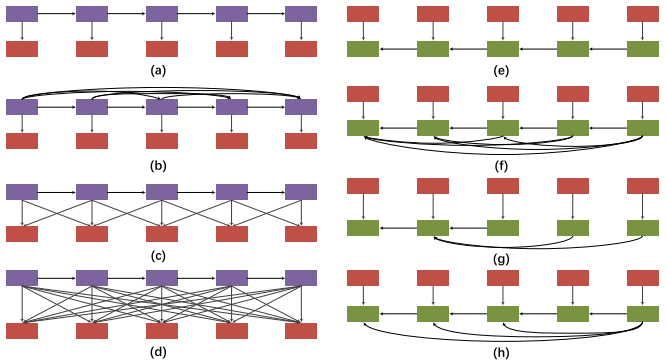


Fig. 1. Illustration of different architectures. The purple block, red block, and green block indicate the encoder, transport layer, and decoder, respectively. Left column: connection patterns between the encoder and the transport layer; right column: connection patterns between the transport layer and the decoder.

fusion difficulties [25]. One of the most significant challenges for ORSI-SOD lies in the great variation of target scales, where some targets may occupy only a few pixel points, while others may be throughout the entire image. Therefore, it seems crucial to fully utilize and aggregate all scale features. However, existing multiscale feature fusion schemes aggregate low-level features containing details and high-level features containing semantics, which leads to the introduction of noise and inaccurate salient mappings. While realizing multiscale feature aggregation, how to reduce the noise interference introduced by inter-layer interactions becomes an issue worthy of in-depth research. In addition, by constructing a global guidance module on the bottom-up decoding path [see Fig. 1(g) and (h)], the location information of potentially salient objects can be provided for different feature layers [26], [27]. However, there still exists the problem that background features may cause interference, leading to inaccurate inferred target locations.

In this case, direct fusion by utilizing means such as attention mechanism may ignore the scale differences between features, leading to poor performance in detecting objects with large-scale variations. In addition, the direct fusion process may introduce irrelevant scale information, increase noise interference, and affect the accuracy of object detection. By analyzing the existing literature [28], [29], SOD methods mainly focus on weighted guidance using low-dimensional texture information to improve the detection performance. However, this approach may not be able to fully characterize the complex relationships between targets at different scales or completely express the rich features of the targets, resulting in limited performance when dealing with scenes with large-scale variations.

To address this challenge, a multiscale indirect fusion method for ORSI-SOD is proposed, that is, a two-stage network with multiscale weighting and multidimensional guidance (WeightNet). Using an indirect full-scale weighed approach, multiscale SMs can be generated in the first stage, and then features of different scales can be weighted in the second stage. This method can better utilize the full-scale information and use the SMs to filter and refine the features, thus reducing noise interference and improving the robustness and accuracy of object detection. Finally, our two-stage network design makes the model more flexible and able to be

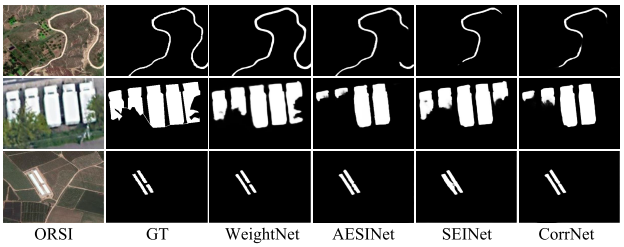


Fig. 2. Visual comparison of our method with three state-of-the-art ORSI-SOD methods including AESINet [19], SEINet [69], and CorrNet [56].

adapted to specific task requirements and data characteristics, which improves the applicability and usefulness of the model. As a result, our method significantly improves the performance and robustness of detection, solves the problems of scale variation and noise interference that exist in existing methods, and demonstrates significant research value and innovation. It is capable of capturing multiscale objects as well as localizing salient objects exactly to generate more accurate SMs, as shown in Fig. 2.

To expand, we deployed two different stages of encoder–decoder networks. The first-stage network is a classical U-Net encoder–decoder network called multiresolution SM extraction network. The second stage network uses the SMs of different resolutions generated by the first-stage network as the guidance information for further feature fusion, which is called weighted feature aggregation network. Then, two key components as indicated in red and blue dashed boxes in Fig. 3 are implemented in the weighted feature aggregation network. The joint action of these components allows the network to obtain multiscale feature representations and accurately localize the position of the target with clear detail information. In addition, we also introduce multiscale parallel branches to the classical tandem decoder block to capture objects of different sizes and increase feature diversity. By the above approach, the proposed WeightNet collects both local and global features and can accurately perceive the local details and present the global context of salient objects.

Our main contributions can be summarized as follows.

1) The WeightNet is proposed, a two-stage network for ORSI-SOD, which not only can fully capture the multiscale information of features without introducing cross-layer feature interaction noise, but also can accurately infer the location and detail information of targets.
2) The multiscale weighted feature aggregation module (MWFAM) is introduced. This module effectively fuses the global and local information of weighted features at multiple scales. In this case, the multiscale weighted matrix is generated from the multiresolution SMs extraction network.
3) Multidimensional feature guidance module (MDFGM) is used to guide the network decoding. This module synthesizes the low-dimensional spatial information and the high-dimensional semantic information of the features to make the final obtained detection results more accurate.
4) In addition, the classic decoder blocks are extended by adding multiscale parallel branches, called multiscale

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:13:46 UTC from IEEE Xplore. Restrictions apply.

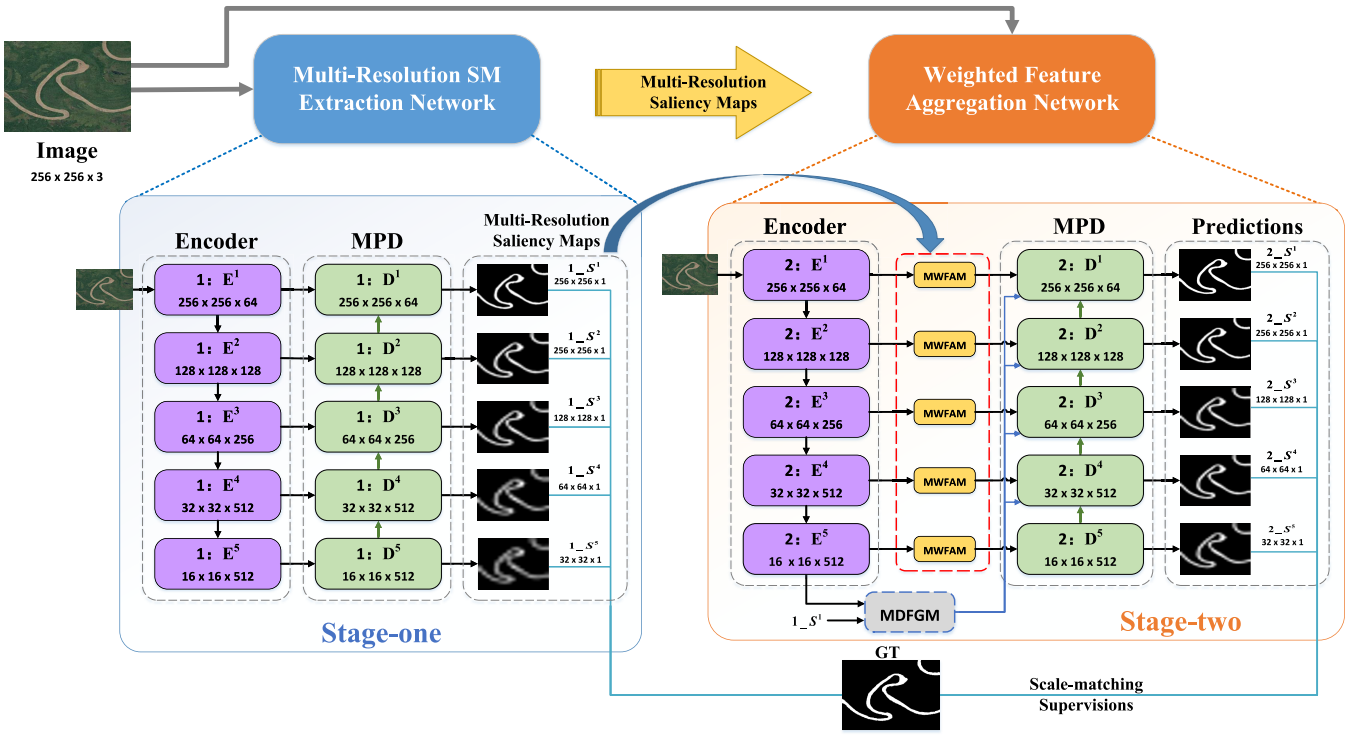


Fig. 3. Overall framework of the proposed WeightNet. First, a multiresolution SM extraction network is executed. We use the classical VGG-16 to extract basic features from ORSI that are 256 × 256 × 3 in size. The basic features are fed directly into the decoder, and deep supervision is used to obtain a multiscale SM. Then, a weighted feature aggregation network is carried out (the weighted information is derived from the first-stage network, as shown by the blue curved arrows in the figure). We model the multiscale global and local information of salient objects in MWFAM. Finally, we utilize the multiscale features of the five MWFAM outputs and gradually infer the salient objects guided by the multidimensional features. In the training phase, we apply scale-matching comprehensive supervision to each MPD in the two-stage network.

parallel decoder (MPD), for increased decoder perception of salient objects at different scales.

II. Related Works

A. Salient Object Detection in Natural Scene Images

Traditional work on SOD for NSI is based on classical principles to generate specific manual features. Itti et al. [30] proposed a salient model of visual attention, laying the foundation for a traditional approach. Liu et al. [31] proposed the method based on kernel density estimation and two-phase graph cut. Liu et al. [32] proposed a new framework focusing on salient region merging and salient node selection. In [33], robust salient detection was performed by regularized random walks ranking. Yuan et al. [34] further build on [33] by combining it with inverse correction. Zou et al. [35] improved the performance by exploring the correlation between salient region detection and target segmentation. Kim et al. [36] extended the approach with high-dimensional color transformations and local spatial support. Zhou et al. [37] used a diffusion process to integrate the diffusion results of a two-layer sparse graph into the final SM. Peng et al. [38] used structured matrix decomposition to detect salient targets. Although traditional methods do not systematically model the human eye visual system, they deeply resolve the connectivity between features and provide constructive solutions.

The CNN-based feature processing strategy has been successfully used for NSI-SOD with excellent performance advantages. Pang et al. [23] integrated features with similar resolution from neighboring encoders and extracted multiscale information using a self-interaction module to improve the representation of targets of different sizes. Huang et al. [39] considered the interactive aggregation of multilayer and multiscale features, integrating features in order to obtain more effective multiscale context information. Zhao et al. [40] designed GateNet, where useful contextual information in the encoder can be optimally transmitted to the decoder. Yan et al. [41] proposed to progressively integrate sparsely labeled and densely labeled new network framework. Zhao et al. [42], in order to solve the rough boundary problem of salient objects, combined the local edge information and global position information through the complementarity between the salient edge information and the salient object information to obtain the salient edge features. Zhou et al. [43] used two independent decoder structures to decode the salient and edge information of the target, respectively, and then fused them for salient target detection. Liu et al. [27] proposed PoolNet based on pooling, which can incrementally refine the high-level semantic features to generate detail-rich SM. Chen et al. [44] proposed a new network framework through the stepwise context-aware feature aggregation, which effectively integrates low-level detail features, high-level semantic features, and global context features. The NSI-SOD methods described above are highly indicative of the ORSI-SOD methods. However, due to the specificity of ORSI, none of these methods directly applied to ORSI-SOD can obtain satisfactory results.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:13:46 UTC from IEEE Xplore. Restrictions apply.

B. Salient Object Detection in Optical Remote Sensing Images

With the booming of NSI-SOD, ORSI-SOD has also received more and more attention. Zhang et al. [45] proposed a two-way complementary saliency analysis scheme that combines vision-oriented saliency analysis and knowledge-oriented saliency analysis. Liu et al. [46] proposed an unsupervised saliency detection model. To avoid the effect of shadows, the CIE Lab color space is used to construct color Markov chains and generate bottom-up potential SMs. Zhang et al. [47] combined superpixel segmentation and information entropy with fusion and enhancement of histogram statistics to generate the final SM. Zhang et al. [48] analyzed the intrinsic connection and contribution between salient features and used this relationship to propose an adaptive multifeature fusion module. Faur et al. [49] used the idea of compression of regions of interest (ROI) for salient object segmentation.

The CNN-based ORSI-SOD solutions achieve superior performance compared to traditional methods. Li et al. [50] proposed an LV-structured nested encoder–decoder network that not only captures the local features of salient objects, but also integrates detailed features and semantic features from different encoder layers. In addition, they provided an ORSI dataset with pixel-level labeling of rich scenes (ORSSD). Zhang et al. [21] extended the ORSSD to provide a more challenging dataset, EORSSD, and also proposed an ORSI-SOD network that densely connects high-level features to low-level features. Zhou et al. [51] used multiscale depth feature fusion and edge cues to generate effective multiscale depth features and edge information for a robust salient object representation. Tu et al. [52] firstly extracted multiscale regional features through an attention module; secondly, combined local and global information to generate boundary features; and finally, introduced boundary features into multiscale features through joint learning to achieve accurate ORSI-SOD. Zhou et al. [28] generated precise boundary information by a separate decoder, which then acted as another encoder to recursively guide the information to calibrate the positional cues of salient targets. Zheng et al. [53] proposed BANet in order to solve the problems of unsmooth edges and incomplete structure in the ORSI-SOD task. Song et al. [54] utilized neighbor complementarity and introduced global constraints to solve the problems of under-representation of neighboring features and improper constraint supervision. In addition, some lightweight ORSI-SOD networks have been proposed for deployment in resource-constrained applications [55], [56], [57].

ORSI-SOD solutions combining CNNs and transformer or transformer-based solutions have also received a lot of attention in recent years. Wang et al. [58] designed a hybrid encoder including a CNN and a transformer to capture global context by unifying the two components to mitigate interference from complex backgrounds. Bao et al. [59] proposed a network based on a two-branch architecture that utilizes an aggregation transformer and CNN to fuse local spatial details and global semantic information into a final high-quality SM. GLGCNet [60] employs a transformer-based encoder to extract global representations and combines them with local context-aware features for comprehensive salient modeling, in addition to an edge assignment module to refine the initial detection. Zhang et al. [61] proposed a transformer-guided dual-flow network, which extracts multiscale features through global receptive fields and refines them according to the characteristics of the feature hierarchy, respectively.

C. Attention Mechanism in Salient Object Detection

Whether in NSI-SOD or ORSI-SOD, we find that the attention mechanism has become a hot topic nowadays and has been introduced to highlight salient regions in many CNN-based methods. For NSI-SOD, Li et al. [62] focused on both foreground and background information and utilized the complementary perceptual attention of the two to enhance the discriminative ability of the network. Huang et al. [63] utilized high-level semantic information for parallel attention fusion to refine the boundaries of salient objects. Chen et al. [64] introduced the attention mechanism in the backbone network using a top-down approach. Zhao and Wu [65] introduced an attention mechanism in pyramid features to highlight the rich contextual features of salient targets. For ORSI-SOD, Li et al. [24] further mined the contextual information of the features using attentional mechanisms to highlight the multiscale properties of salient targets. Li et al. [20] captured multiple features (foreground, background, boundary, and global features) of the target simultaneously using multiple attention mechanisms. Gu et al. [66] addressed the loss of attention prevalent in the single use of spatial attention and channel attention by using 2-D attention and full-stage semantic guidance. Zheng et al. [67] proposed CIFNet, which aggregates multiscale information through globally guided attention that interacts across scales.

Prominent objects in ORSI usually have complex geometries, texture features, and different sizes and orientations, and are often accompanied by occlusions, shadows, and nonuniform illumination. The approaches described above present meaningful solutions to the characterization of ORSI. However, we believe that the multiscale information in ORSI needs to be further explored, which is important to bridge these challenging scenarios.

III. Proposed Method

A. Network Overview

The general framework diagram of WeightNet is shown in Fig. 3. It consists of encoder–decoder networks, MWFAM, MDFGM, and MPD. Both of the two stages of the backbone in this network use the classical encoder–decoder network paradigm. In this, the multiresolution SM extraction network (first-stage network) performs the initial extraction of SMs at multiple resolutions. The second stage network, called the weighted feature aggregation network, uses the SMs at different scales obtained from the first-stage network as the weighted information for further refinement in this stage. The weighted feature aggregation network fully fuses features from different scales to further explore a more adequate multiscale information. In addition, multidimensional feature

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:13:46 UTC from IEEE Xplore. Restrictions apply.

guidance module is introduced that more deeply integrates low-dimensional detail information with high-dimensional semantic information. This module not only captures the exact location of salient objects more accurately, but also sharpens their details. At the same time, the decoder is further optimized to improve the ability to infer multiscale information of features.

B. Encoder

Since the proposed WeightNet is independent of a specific type of backbone architecture and does not lose generality, we use the commonly used VGG-16 architecture to describe our approach. The encoder for both stages of the network uses the same VGG-16 architecture for the original feature extraction. Unlike the original VGG-16 for image classification task, we removed the last four layers, including a maximum pooling layer and three fully connected layers. For our pixel-level ORSI-SOD task, the remaining convolutional blocks are denoted as $n : E^t$, where n is the stage index, $n = \{1, 2\}$; t is the block index, belonging to $\{1, 2, 3, 4, 5\}$. For $t = 1, 2$, $n : E^t$ contains two convolutional layers; for $t = 3, 4, 5$, $n : E^t$ contains three convolutional layers. For the input optical remote sensing images (ORSIs) $I \in \mathbb{R}^{256\times256\times3}$, the features extracted by each convolutional block are denoted as $n_f^t \in \mathbb{R}^{H_t\times W_t\times C_t}$, where H_t is $(256/2^{t-1})$, W_t is $(256/2^{t-1})$, C_t is $\{64, 128, 256, 512, 512\}$. The original features generated by encoder of the first-stage network are directly fed to decoder. The multiscale SMs are generated by deep supervision and are fed to the second stage network. The original features generated by the encoder of the second stage network are fed into the MWFAM to generate new augmented features.

C. Multiscale Parallel Decoder

Decoder networks are responsible for inferring salient targets. In general, a classical decoder network consists of five common decoder blocks, where each decoder block is a cascade of three 3×3 convolutional layers [20], [28]. However, the underlying cascade structure with a single convolutional kernel does not do a good job of capturing salient objects with varying sizes or uncertain quantities. This leads to insensitivity to multiscale targets in unique scenes of ORSI, which may compromise the inference accuracy of salient objects in the decoder network. As mentioned earlier, using convolutional layers with different convolutional kernels in parallel is a popular strategy for capturing both local and global content within a feature level. We add a parallel dilated 3×3 convolutional layer to the convolution of each decoder block and sum the features of the different convolutional kernels of that layer for feature fusion, as shown in Fig. 4. This approach increases the ability of the decoder to capture multiscale targets. We define the processing of the MPD as $D(\cdot)$ with the following equation:

$$\hat{f}_{\text{out}}^n = D\left(\hat{f}_{\text{in}}^n\right) \tag{1}$$

$$D\left(\hat{f}_{\text{in}}^n\right) = \text{conv}^3\left(\text{conv}^2\left(\text{conv}^1\left(\hat{f}_{\text{in}}^n\right)\right)\right) \tag{2}$$

$$\text{conv}^n(\mathcal{L}) = r^n(\mathcal{L}) + \text{conv}_{3\times3}(\mathcal{L}) \tag{3}$$

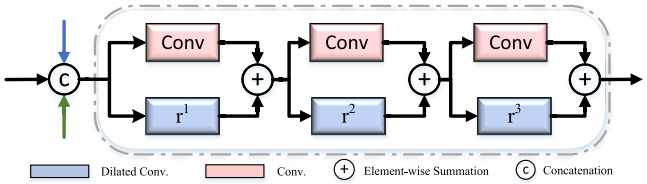


Fig. 4. Illustration of the MPD.

TABLE I
DETAILED PARAMETERS OF THREE PARALLEL BRANCHES (I.E., DILATED CONVOLUTIONS) IN MPD, INCLUDING KERNEL SIZE, DILATION RATE, AND CHANNEL NUMBER

Aspects	Dilated conv	r^1	r^2	r^3	Channel Number
$1 : D^1/2 : D^1$	3×3	5	3	2	64
$1 : D^2/2 : D^2$	3×3	5	3	2	123
$1 : D^3/2 : D^3$	3×3	5	3	2	256
$1 : D^4/2 : D^4$	3×3	3	2	2	512
$1 : D^5/2 : D^5$	3×3	3	2	2	512

where, $\hat{f}_{\text{in}}^n$ denotes the input features of the decoder block, and if there is more than one input feature, they are used as input features after the concatenation operation. $\hat{f}_{\text{out}}^n$ denotes the output features of the decoder block. $\text{conv}^n(\cdot)$ is the convolutional operation of the nth convolutional layer of the decoder block, $n = 1, 2, 3$. $\mathcal{L}$ is the input feature of the convolutional operation. The operation process of $r^n(\cdot)$ includes three processes: dConv (3×3 dilated convolutional layer), BN (batch normalization layer), and ReLU; the operation process of $\text{conv}_{3\times3}(\cdot)$ involves three processes: Conv (3×3 convolutional layer), BN, and ReLU. Between the two decoder blocks, an inverse convolutional layer is used to align the resolution. The MPD is denoted as $n : D^t$, where n is the stage index, $n = \{1, 2\}$; t is the block index, belonging to $\{1, 2, 3, 4, 5\}$, as shown in Fig. 3. The same structure is used for the decoder in both stages of the network, and the dilation coefficients corresponding to the different dilation convolutional layers are shown in Table I.

D. Multiscale Weighted Feature Aggregation Module

For multiscale feature fusion, existing methods usually have the aggregation of different spans of low-level features containing details and high-level features containing semantics, leading to the introduction of noise in the aggregated features and producing inaccurate saliency mappings. The multiscale weighted feature aggregation module (MWFAM) not only adds multiscale information to the original features, but also does not add information about noise that may be generated by the fusion of features from different layers. This module is connecting the encoder and decoder in the second stage network as shown in the red dashed box in Fig. 3.

The specific design of this module is shown in Fig. 5. We perform weighted fusion for the initial features 2_f^t generated by the encoder of the second stage network. First, it is necessary to change the size of different scales of SMs generated by the first-stage network and stack them to generate weighted data blocks matching the original features that need to be weighted. Since the different scales of SM produced by

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:13:46 UTC from IEEE Xplore. Restrictions apply.

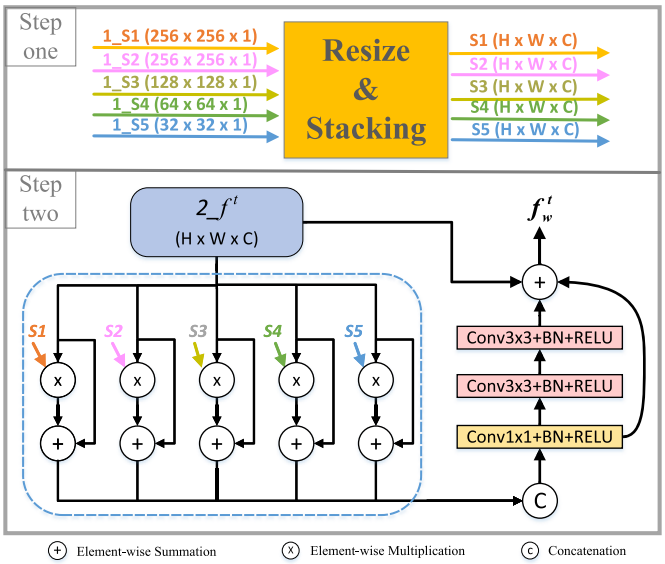


Fig. 5. Illustration of the MWFAM.

the first-stage network are the result of deep supervision, they have the same number of channels and GT, which are all 1, but the scales are not the same. In order to weight the original features, we upsample or downsample SMs of different sizes, resize them to be the same size as the original features, and then perform feature stacking so that the number of channels is the same as the original features of the layer. The processing equations for SM resize and stacking are as follows:

$$S^t_{\text{resize}} = \text{Sample}(1_s^t) \tag{4}$$

$$s^t = \text{concat}\left(\underbrace{S^t_{\text{resize}}, S^t_{\text{resize}}, \ldots, S^t_{\text{resize}}}_{C}\right) \tag{5}$$

where, S^t_{resize} is the SM after resizing, $t = \{1, 2, 3, 4, 5\}$; 1_s^t is the SM of the different scales obtained for the first-stage network. Sample$(\cdot)$ is the sampling operation, where upsampling is done using the bilinear method and downsampling is done using the maxpool method. s^t is the result after SM stacking; concat is the concat operation; C is the number of channels of the original feature being weighted.

Then, multiplication and addition operations are performed on the original features with SMs of different scales. After that, concat operation is performed on the weighted features. Next, a residual block is used for local feature extraction. Specifically, since the number of channels of the concatenated features becomes five times of the original features. At first, a 1×1 convolution is used for feature dimensionality reduction to make it the same number of channels as the original feature; then a residual block of two layers of 3×3 convolution is used to obtain features that incorporate different scales, local and global information. Finally, one addition operation is performed with the original features as the final weighted feature to be fed into the corresponding decoder block. The specific operation flow is shown in the following equation:

$$f^t = E^t \times s^t + E^t \tag{6}$$

$$\hat{f}^t = \text{concat}(f^1, f^2, f^3, f^4, f^5) \tag{7}$$

$$\hat{f}^n_{1\times1} = \text{Conv}_{1\times1}(\hat{f}^t) \tag{8}$$

$$\hat{f}^t_{\text{res}} = \text{Conv}_{3\times3}\left(\text{Conv}_{3\times3}\left(\hat{f}^n_{1\times1}\right)\right) + \hat{f}^n_{1\times1} \tag{9}$$

$$f^t_w = \hat{f}^t_{\text{res}} + E^t \tag{10}$$

where, $t = 1, 2, 3, 4, 5$; E^t denotes the original features generated by different encoder blocks of the second stage network. f^t is the result of the original features weighted by multiple proportions. s^t is the weight block of SM after the resize and stacking operation. $\hat{f}^t$ is a fusion feature of multiscale weighted features; $\hat{f}^t_{1\times1}$ is the feature after dimensionality reduction by 1×1 convolution; $\hat{f}^t_{\text{res}}$ is the feature after a two-layer 3×3 convolutional residual block; f^t_w is the final output weighted feature of the MWFAM. concat is the concat operation. The operation process of $\text{Conv}_{1\times1}$ includes three processes: Conv (1×1 convolutional layer), BN (batch normalization layer) and ReLU. The operation process of $Conv_{3\times3}$ includes three processes: Conv (3×3 convolutional layer), BN (batch normalization layer), and ReLU.

E. Multidimensional Feature Guidance Module

As known from the article PoolNet [27], high-dimensional semantic information is gradually diluted during the decoding process due to upsampling, and ultimately inaccurate location information is obtained. In order to solve this problem, a location guidance module is proposed in PoolNet. On this basis, we do further optimization. By adding the weighted detail information of the saliency regions as a guider, we can not only get the accurate position information, but also get more accurate detail and edge information.

The design of the MDFGM is shown specifically in Fig. 6. In the first place, a 3×3 convolution operation is done on the weighted semantic feature of the highest layer for further feature extraction and dimensionality reduction, which is done to reduce the complexity of the network. The feature is used as the baseline for upsampling as the location guidance information for the decoding block. In the decoding process, the position information is as important as the detail information. In order to retain the location information while highlighting the detail information, we adopt the SM containing the most detail information obtained from the deep supervision of the network in the first stage to weight the features. Similarly, in order to match the size and the number of channels of the features, the SM first needs to be resized and then stacking processed as in Section III-D. After the multiplication and addition operation, feature fusion and dimensionality reduction are performed through convolution, as the final guidance information fed into different decoder blocks. The processing of the multidimensional feature guidance module can be expressed as follows:

$$\hat{f}^n_{\text{guide}} = \text{Conv}_{3\times3}\left(\text{up}\left(\text{Conv}_{3\times3}\left(f^5_w\right)\right) \times \text{down}\left(s^1\right) + \text{up}\left(\text{Conv}_{3\times3}\left(f^5_w\right)\right)\right) \tag{11}$$

where, $\hat{f}^n_{\text{guide}}$ is the guidance feature for different decoder blocks, $n = 1, 2, 3, 4$; f^5_w is the weighted semantic feature of the encoder at the top layer; s^1 is the result after stacking the SM in the lowest dimension of the first-stage network; up is the upsampling using bilinear method and down is the downsampling using maxpool.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:13:46 UTC from IEEE Xplore. Restrictions apply.

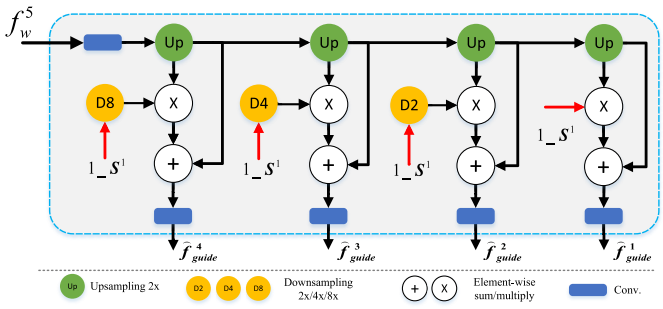


Fig. 6. Illustration of the MDFGM.

F. Loss Function

With a good deep learning network, an effective architecture and several adapted modules are needed. In this way, a good training strategy can improve the efficiency of the model without the need of additional model parameters. We employ the widely used deep supervision in the training phase to monitor intermediate SMs of different sizes, forcing the features to learn the features of saliency regions of different sizes. We employ the classical pixel-level BCE loss, graph-level IoU loss, and metric-aware F-m loss [20] in the loss function for network training, which can be formulated as

$$L_n^t = l_{\text{bce}}(\text{up}(n_s^t), \text{GT}) + l_{\text{iou}}(\text{up}(n_s^t), \text{GT}) + l_{\text{fm}}(\text{up}(n_s^t), \text{GT}) \tag{12}$$

where, GT stands for ground truth; n_s^t is the SM of the tth decoded block of the nth stage of WeightNet. $l_{\text{bce}}(\cdot)$, $l_{\text{iou}}(\cdot)$, and $l_{\text{fm}}(\cdot)$ are BCE loss, IoU loss, and F-m loss, respectively. We denote the total loss function $\mathcal{L}_{\text{total}}$ of WeightNet as

$$\mathcal{L}_{\text{total}} = \sum_{n=1}^{2}\sum_{t=1}^{5} L_n^t. \tag{13}$$

IV. Experiment

A. Datasets and Implementation Details

1) Datasets: To have a comprehensive validation of our model, we have trained and tested by using three public ORSI datasets. These datasets are ORSSD dataset [50], EORSSD dataset [21], and ORSI-4199 [52] dataset. To be more specific, the ORSSD dataset contains 800 images, of which 600 images were used as the training set and 200 images were used as the testing set. This dataset contains different spatial resolutions, different target object scales and types, and different scene images. EORSSD is an extension of ORSSD, regardless of the image scale, type, and complex scene. In particular, EORSSD contains 2000 images, of which 1400 images are used for training and 600 images are used for testing. The ORSI-4199 dataset, which includes more challenging scenarios, contains 4199 images, of which 2000 are used for training and 2199 for testing. Most importantly, each image in ORSSD, EORSSD, and ORSI-4199 datasets provides pixel-level GTs, which facilitates us to have a fair evaluation criterion.

2) Implementation Details: In order to make WeightNet more robust and prevent model overfitting, we augment the training set by rotating it at angles of 90°, 180°, and 270°, as well as mirroring reflections of these images before rotating them at angles of 90°, 180°, and 270°. This approach can expand the training sets of ORSSD, EORSSD, and ORSI-4199 to 4800 examples, 11 200 examples, and 16 000 examples, respectively. In addition, each training image is resized to a fixed size of 256 × 256 during the training phase.

We carried out experiments on WeightNet on the PyTorch platform with an NVIDIA Tesla V100 (32 GB RAM). The parameters of the pre-trained VGG-16 model were used to initialize the parameters of the encoder network in WeightNet. In addition, the Adam optimization strategy was used for WeightNet. The batch size was 8; the initial learning rate was e^{-4}, divided by 10 after 40 epochs; the maximum number of epochs was set to 60.

B. Evaluation Metrics

For a more comprehensive quantitative comparison of different significance models, we use ten widely used evaluation metrics to evaluate our method compared to other comparative methods. These metrics include S-measure ($S_\alpha, \alpha = 0.5$), F-measure ($F_\beta^{\max}, F_\beta^{\text{mean}}, F_\beta^{\text{adp}}, \beta^2 = 0.3$), E-measure ($E_\xi^{\max}, E_\xi^{\text{mean}}, E_\xi^{\text{adp}}$), mean absolute error (MAE, $\mathcal{M}$), precision–recall (PR) curve and F-measure curve. Among them, S-measure measures the structural similarity of the SMs, which incorporates both region similarity and object similarity. F-measure is considered as a composite metric, which can be obtained by weighted harmonic averaging of precision and recall. E-measure evaluates the predicted SMs by combining both the local pixel saliency values and the image-level average saliency values. MAE gives the absolute pixel-level difference between the SMs and their corresponding GTs. MAE gives the absolute pixel-level difference between a significance mapping and its corresponding GT. PR curve is a standard metric for assessing model performance. In particular, larger values of S-measure, F-measure, and E-measure are better and MAE is smaller. The closer the PR curve is to the coordinate (1, 1), the better the performance of the significance model. The larger the area of the coordinate covered by the F-measure curve, the better the performance of the significance model.

C. Comparison With State-of-the-Art Methods

In this section, the proposed WeightNet and current state-of-the-art methods are analyzed in both quantitative and qualitative details. To make a fair comparison, we used the publicly available pre-trained models of the comparison methods or the SMs provided by the authors for the calculations. Since the CNN-based NSI-SOD methods were not trained and tested on ORSI datasets, we retrained the methods with default parameter settings consistent with ours. Furthermore, since some methods use different backbones, the quantitative analysis below will provide a relatively fair comparison and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:13:46 UTC from IEEE Xplore. Restrictions apply.

TABLE II

QUANTITATIVE COMPARISON OF THE WEIGHTNET WITH STATE-OF-THE-ART METHODS ON THE EORSSD AND ORSSD DATASETS. ↑ MEANS BIGGER IS BETTER AND ↓ MEANS SMALLER IS BETTER. THE TOP TWO RESULTS ARE DENOTED IN **RED** AND **BLUE**, RESPECTIVELY

Methods	Backbone	#Param (M) ↓	FLOPs (G) ↓	EORSSD Dataset								ORSSD Dataset							
				S_α ↑	F_β^{max} ↑	F_β^{mean} ↑	F_β^{adp} ↑	E_ξ^{max} ↑	E_ξ^{mean} ↑	E_ξ^{adp} ↑	$\mathcal{M}$ ↓	S_α ↑	F_β^{max} ↑	F_β^{mean} ↑	F_β^{adp} ↑	E_ξ^{max} ↑	E_ξ^{mean} ↑	E_ξ^{adp} ↑	$\mathcal{M}$ ↓
CNN-based NSI-SOD methods																			
R3Net [68]	VGG16	56.16	47.5	.8184	.7498	.6302	.4165	.9483	.8294	.6462	.0171	.8141	.7456	.7383	.7379	.8913	.8681	.8887	.0399
PoolNet [27]	VGG16	53.63	123.4	.8207	.7545	.6406	.4611	.9292	.8193	.6836	.0210	.8403	.7706	.6999	.6166	.9343	.8650	.8124	.0358
EGNet [42]	ResNet50	108.07	291.9	.8601	.7880	.6967	.5379	.9570	.8775	.7566	.0110	.8721	.8332	.7500	.6452	.9731	.9013	.8226	.0216
MINet [23]	VGG16	47.56	146.3	.9040	.8344	.8174	.7705	.9442	.9346	.9243	.0093	.9040	.8761	.8574	.8251	.9545	.9454	.9423	.0144
GateNet [40]	ResNet50	100.02	108.3	.9114	.8566	.8228	.7109	.9610	.9385	.8909	.0095	.9186	.8871	.8679	.8229	.9664	.9538	.9428	.0137
PA-KPN [69]	ResNet50	141.06	617.7	.9192	.8639	.8358	.7993	.9616	.9536	.9416	.0104	.9239	.8890	.8727	.8548	.9680	.9620	.9579	.0139
CNN-based ORSI-SOD methods																			
LVNet [50]	VGG16	-	-	.8630	.7794	.7328	.6284	.9254	.8801	.8445	.0146	.8815	.8263	.7995	.7506	.9456	.9259	.9195	.0207
DAFNet [21]	VGG16	29.35	68.51	.9166	.8614	.7845	.6427	**.9861**	.9291	.8446	**.0060**	.9191	.8928	.8511	.7876	.9771	.9539	.9360	.0113
SARNet [63]	VGG16	25.91	129.7	.9240	.8719	.8541	.8304	.9620	.9555	.9536	.0099	.9134	.8850	.8619	.8512	.9557	.9477	.9464	.0187
MJRBM [52]	VGG16	43.54	95.7	.9197	.8656	.8239	.7066	.9646	.9350	.8897	.0099	.9204	.8842	.8566	.8022	.9623	.9415	.9328	.0163
EMFINet [51]	VGG16	107.26	480.9	.9260	.8720	.8486	.7984	.9711	.9604	.9501	.0084	.9366	.9002	.8856	.8617	.9737	.9671	.9663	.0109
ERPNet [28]	VGG16	56.48	113.71	.9210	.8632	.8304	.7554	.9603	.9401	.9228	.0089	.9254	.8974	.8745	.8356	.9710	.9566	.9520	.0135
ACCoNet [24]	VGG16	102.55	184.27	.9290	.8837	.8552	.7969	.9727	.9653	.9450	.0074	.9437	.9149	.8971	.8806	.9796	**.9754**	.9721	.0088
CorrNet [57]	VGG16	4.09	21.1	.9289	.8778	.8620	.8311	.9696	.9646	.9593	.0083	.9380	.9129	**.9002**	.8875	.9790	.9746	.9721	.0098
SeaNet [56]	MobileNet	2.76	1.7	.9208	.8649	.8519	.8304	.9710	.9651	.9602	.0073	.9260	.8942	.8772	.8625	.9767	.9722	.9670	.0105
SEINet [70]	VGG16	20.16	50.1	.9283	.8788	**.8708**	**.8587**	.9723	**.9672**	**.9657**	.0076	.9382	.9090	.8996	**.8946**	.9784	.9746	**.9763**	.0097
AESINet [19]	VGG16	41.22	110.8	**.9358**	**.8838**	.8524	.7923	.9751	.9636	.9462	.0079	**.9460**	**.9193**	.8986	.8666	**.9828**	.9747	.9707	**.0086**
BSCGNet [29]	VGG16	26.99	86.35	.9294	.8807	.8664	**.8504**	.9692	.9649	.9631	.0079	.9411	.9101	.8978	.8897	.9784	.9745	.9737	.0093
Ours-VGG	VGG16	99.05	221.46	**.9384**	**.8907**	**.8764**	**.8472**	**.9756**	**.9710**	**.9659**	**.0063**	**.9481**	**.9188**	**.9072**	**.8971**	**.9817**	**.9779**	**.9787**	**.0075**
Ours-Res	ResNet50	120.71	66.09	**.9380**	**.8885**	**.8731**	**.8409**	**.9782**	**.9729**	**.9631**	**.0060**	**.9478**	**.9200**	**.9048**	**.8911**	**.9807**	**.9766**	**.9719**	**.0082**
Ours-PVT	PVT	66.34	21.18	**.9417**	**.8908**	**.8779**	**.8521**	**.9809**	**.9749**	**.9670**	**.0060**	**.9512**	**.9278**	**.9125**	**.8979**	**.9864**	**.9823**	**.9776**	**.0075**
Ours-Hyb	PVT & VGG16	103.65	143.9	**.9427**	**.8955**	**.8800**	**.8437**	**.9808**	**.9760**	**.9674**	**.0053**	**.9536**	**.9281**	**.9159**	**.9043**	**.9863**	**.9826**	**.9801**	**.0068**

TABLE III

QUANTITATIVE COMPARISON OF THE WEIGHTNET WITH STATE-OF-THE-ART METHODS ON THE ORSI-4199 DATASETS. ↑ MEANS BIGGER IS BETTER AND ↓ MEANS SMALLER IS BETTER. THE TOP TWO RESULTS ARE DENOTED IN **RED** AND **BLUE**, RESPECTIVELY

Methods	ORSI-4199 Dataset							
	S_α ↑	F_β^{max} ↑	F_β^{mean} ↑	F_β^{adp} ↑	E_ξ^{max} ↑	E_ξ^{mean} ↑	E_ξ^{adp} ↑	$\mathcal{M}$ ↓
CNN-based NSI-SOD methods								
R3Net [68]	.8142	.7847	.7790	.7776	.8880	.8722	.8645	.0401
PoolNet [27]	.8271	.8010	.7779	.7382	.8964	.8676	.8531	.0974
EGNet [42]	.8464	.8267	.8041	.7650	.9161	.8947	.8620	.0440
PA-KRN [69]	.8491	.8415	.8324	.8200	.9280	.9168	.9063	.0382
DPORTNet [71]	.8094	.7789	.7701	.7554	.8759	.8687	.8628	.0569
DNTD [72]	.8444	.8310	.8208	.8065	.9158	.9050	.8963	.0425
CNN-based ORSI-SOD methods								
MJRBM [52]	.8593	.8493	.8309	.7995	.9311	.9102	.8891	.0374
EMFINet [51]	.8675	.8584	.8479	.8186	.9340	.9257	.9136	.0330
ERPNet [28]	.8670	.8553	.8374	.8024	.9290	.9149	.9024	.0357
ACCoNet [24]	**.8775**	.8686	.8620	.8581	.9412	.9342	.9167	.0314
MCCNet [20]	.8746	**.8690**	**.8630**	**.8592**	.9413	**.9348**	.9182	.0316
CorrNet [57]	.8623	.8560	.8513	.8534	.9330	.9206	.9142	.0366
SeaNet [56]	.8722	.8653	.8591	.8556	**.9426**	**.9363**	**.9197**	**.0308**
SEINet [70]	.8686	.8616	.8572	.8571	.9369	.9305	.9138	.0330
AESINet [19]	.8702	.8670	.8573	.8251	.9403	.9310	**.9184**	.0325
Ours-VGG	**.8786**	**.8695**	**.8633**	**.8590**	**.9420**	**.9330**	**.9157**	**.0298**
Ours-Res	**.8820**	**.8749**	**.8664**	**.8569**	**.9459**	**.9359**	**.9188**	**.0293**
Ours-PVT	**.8884**	**.8870**	**.8813**	**.8802**	**.9534**	**.9456**	**.9283**	**.0267**
Ours-Hyb	**.8912**	**.8873**	**.8804**	**.8756**	**.9540**	**.9486**	**.9276**	**.0260**

analysis of our proposed network using backbone VGG-16 (Ours-VGG) versus methods using the same backbone.

1) Quantitative Comparison: We report the quantitative measurements of EORSSD and ORSSD by the proposed method and all comparative methods in Table II. On the whole, our method demonstrates excellent performance on both public datasets, outperforming all the methods compared. Specifically, on the EORSSD dataset, all five metrics (S_α, $F_\beta^{\max}$, F_β^{mean}, E_ξ^{mean}, E_ξ^{adp}) of WeightNet are optimal and dominant. For S_α and $F_\beta^{\max}$, WeightNet's values are 0.28% and 0.78% higher than the second-place AESINet, respectively. With respect to F_β^{mean}, F_ξ^{mean}, and E_ξ^{adp}, our method outperforms the second-ranked SEINet by 0.64%, 0.39%, and 0.02%, respectively. WeightNet is suboptimal with two metrics ($E_\xi^{\max}$ and $\mathcal{M}$) below DAFNet, but the other six metrics are much higher than DAFNet. On the ORSSD dataset, our method is superior to all compared methods on six metrics. Concretely, for S_α and $\mathcal{M}$, our method is superior to the second-ranked method by 0.22% and 0.0011%, respectively. For F-measure indicators including F_β^{mean} and F_β^{adp}, our method beats the second-ranked method by 0.78% and 0.25%, respectively. In the case of E-measure indicators including E_ξ^{mean} and E_ξ^{adp}, our method is 0.26% and 0.68% higher than the second-place method, respectively. Compared with traditional methods, our method is far ahead. Although CNN-based NSI-SOD methods are retrained on ORSI datasets, their performance is usually lower than that of specialized ORSI-SOD methods, which illustrates the urgency and necessity of proposing specialized solutions for ORSI.

We report the quantitative measurements of ORSI-4199 by the proposed methods and all comparative methods in Table III. Overall, our method performs equally well on the ORSI-4199 dataset, outperforming all compared methods. Specifically, four metrics (S_α, $F_\beta^{\max}$, F_β^{mean}, and $\mathcal{M}$) of WeightNet are optimal, being 0.13%, 0.06%, 0.03%, and.00016 higher than the second-place method, respectively. For F_β^{adp} and $F_\xi^{\max}$, WeightNet's values are suboptimal, being 0.02% and 0.06% lower than the second-place AESINet, respectively.

We can also see from Tables II and III that our CNN-based approach has limited improvement in F-measure and E-measure metrics, although it has some advantages. This is mainly due to the fact that while CNN-based models are able to evaluate the predicted SMs when dealing with salient detection tasks, their limitations lie in the relatively weak global information capture and long-range dependency modeling capabilities. This means that when dealing with complex salient targets or scenes, CNN-based methods may not be able to fully understand the overall semantic and contextual information of the image, resulting in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:13:46 UTC from IEEE Xplore. Restrictions apply.

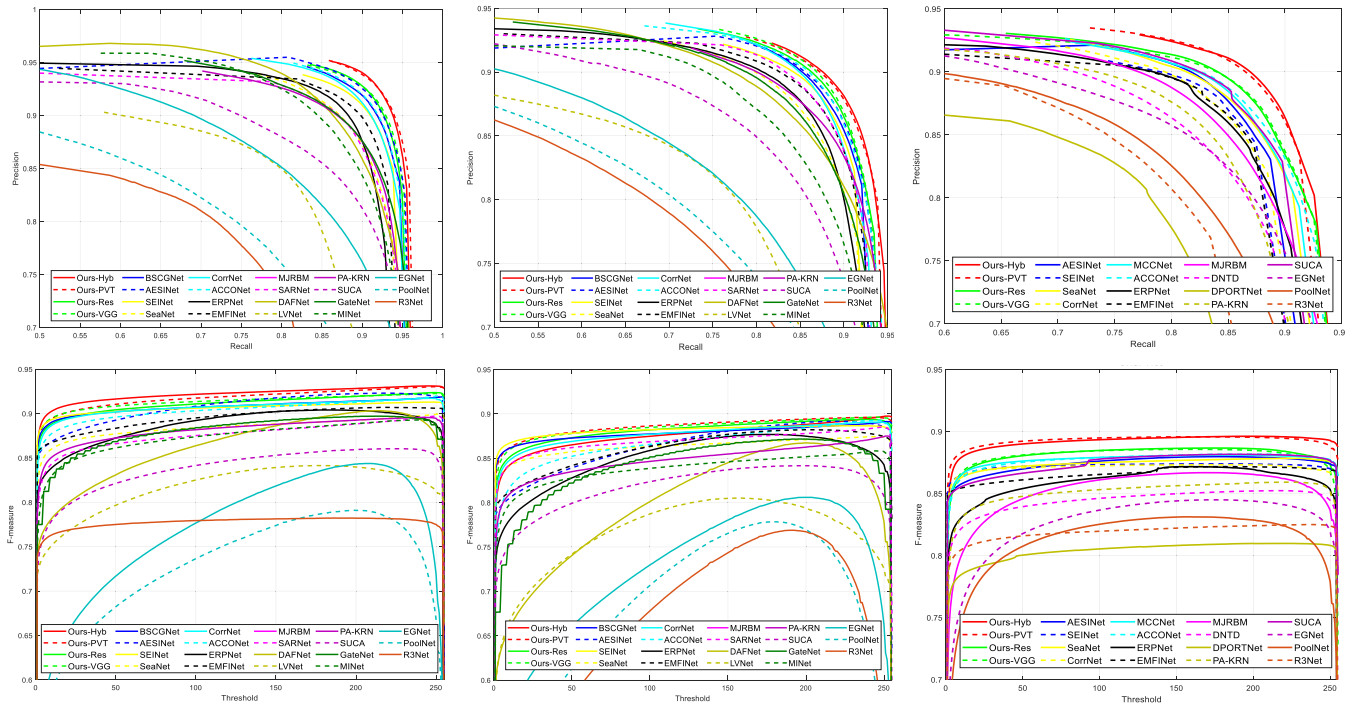


Fig. 7. Quantitative comparison of PR curves and F-measure curves of different salient models on three datasets. The first row shows the PR curves. The second row is the F-measure curve. Each row from left to right represents the results of ORSSD, EORSSD, and ORSI-4199, respectively.

TABLE IV

ATTRIBUTE-BASED PERFORMANCE COMPARISON RESULTS ON ORSI-4199 DATASET. S-MEASURE SCORES ARE GIVEN FOR SPECIFIC ATTRIBUTES. THE LAST COLUMN REPORTS THE AVERAGE RESULTS FOR NINE ATTRIBUTES. THE TOP TWO RESULTS ARE DENOTED IN **RED** AND **BLUE**, RESPECTIVELY

Methods	BSO	CS	CSO	ISO	LCS	MSO	NSO	OC	SSO	Avg.
MJRBM [52]	.9029	.8846	.8826	.8560	.8290	.8250	.8626	.8406	.8350	.8576
EMFINet [51]	.9180	.8909	.8905	.8719	.8399	.8456	.8790	.8304	.8339	.8667
ERPNet [28]	.9057	.8888	.8772	.8490	.8369	.8424	.8864	.8358	.8416	.8526
ACCoNet [24]	.9167	.8988	**.8918**	**.8726**	.8443	**.8585**	.8828	.8610	**.8609**	**.8763**
MCCNet [20]	.9151	**.9012**	.8846	.8682	.8408	.8531	**.8925**	**.8672**	.8525	.8750
CorrNet [57]	.8903	.8858	.8641	.8401	.8331	.8495	.8768	.8582	.8496	.8608
SeaNet [56]	**.9186**	.8982	**.8994**	.8696	.8433	.8442	.8876	.8505	.8449	.8729
SEINet [70]	.9137	.8952	.8737	.8704	.8393	**.8580**	.8764	.8468	.8473	.8689
AESINet [19]	.9091	.8959	.8796	.8630	**.8456**	.8433	.8847	.8414	.8418	.8671
Ours-VGG	**.9234**	**.8994**	**.8917**	**.8827**	**.8450**	**.8571**	**.8946**	**.8676**	**.8572**	**.8794**

limited performance, especially when evaluating E-measure. In contrast, transformer-based models and hybrid models can significantly improve F-measure and E-measure metrics. This is because the transformer model has excellent global perception and long-range dependency modeling capabilities, which can better capture the global semantic information in the image, thus improving the accuracy and robustness of salient detection.

In Fig. 7, we plot the PR curves and F-measure curves for WeightNet and all compared methods. The PR curves of our proposed method (including all variants) are closest to the upper right on all three datasets. Moreover, the F-measure curve covers the largest area on all three datasets. It can be concluded that WeightNet obtains better performance than all the compared models.

2) Attribute Analysis on ORSI-4199: The ORSI-4199 dataset provides nine challenging scenario test datasets for a more comprehensive comparison. To ensure relatively fair evaluation criteria, we only provide the VGG version for comparison. In Table IV, we present the S-measure scores of WeightNet (Ours-VGG) compared with the nine SOTA ORSI-SOD methods. The results show that the proposed model ranks first in four out of nine attributes. In addition, our model ranks first overall in terms of average performance. These results indicate that WeightNet outperforms existing models in most challenging scenarios.

Specifically, WeightNet shows significant advantages in scenarios with big salient object (BSO) and incomplete salient object (ISO). This may be attributed to the multiscale weighed and multidimensional guidance mechanism in our model, which helps to distinguish the target from the background. WeightNet also shows advantages in scenes with narrow salient object (NSO) and off-center (OC). This reflects our model's ability to process edge information, allowing it to localize the boundaries of the target more accurately. Although our model did not score at the highest level in complex scenes (CSs) and low contrast scenes (LCSs), it still showed good robustness. This may be attributed to the feature weighted and filtering mechanisms, which help reduce noise interference and improve the stability of target detection. In addition, our model also has some advantages under the conditions of small salient objects (SSOs), complex salient objects (CSOs), and multiple salient objects (MSOs). This indicates that our model can effectively handle diverse target shapes and sizes and maintain high detection accuracy in multitarget scenarios.

In summary, WeightNet demonstrates advantages in different scene modes, reflecting its multifaceted capabilities in dealing with different problems, including adaptability to target size, shape, location, and scene complexity, as well as resistance to interference and noise.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:13:46 UTC from IEEE Xplore. Restrictions apply.

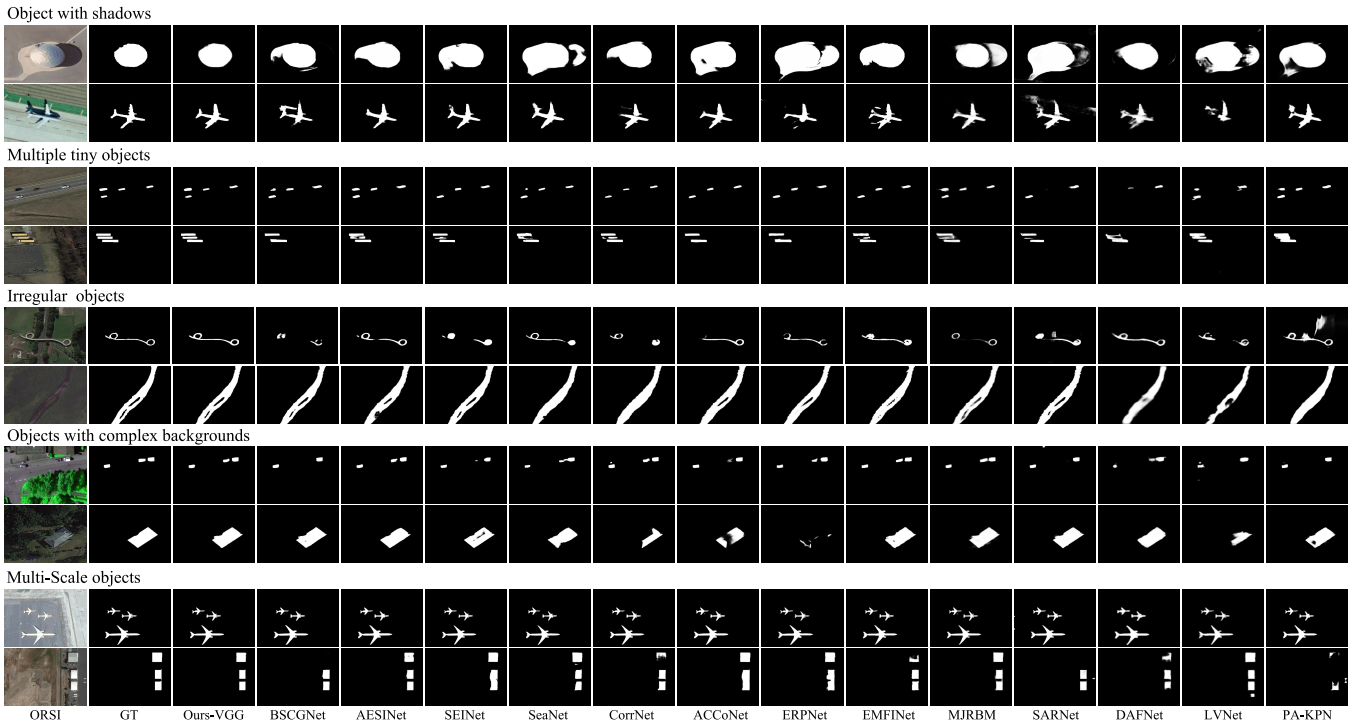


Fig. 8. Visual comparison with 13 representative state-of-the-art methods. The zoomed-in view will be best for perception.

3) Flexibility of Our Method: To validate the flexibility of our approach, we provide three variants of different backbone networks (Ours-Res, Ours-PVT, and Ours-Hyb, respectively) and report their performance on different datasets in Tables II and III. Among them, Ours-Res and Ours-PVT are consistent with our original method (i.e., Ours-VGG), and the same feature extraction network is used in both stages. In particular, the backbone network of Ours-Res is ResNet50, and the backbone network of Ours-PVT is pyramid vision transformer (PVT-v2-b2). To further illustrate the flexibility of our proposed approach, we provide a variant (Ours-Hyb) that employs different encoders for the two stages. It uses PVT for the encoder of the first stage and VGG-16 for the encoder of the second stage.

The results show that, for Ours-Res, its performance advantage becomes more and more obvious as the dataset size increases. This may be attributed to the fact that ResNet50 has a more powerful feature characterization capability when dealing with complex datasets, allowing the model to better capture the salience information of the target. As for Ours-PVT, it shows higher potential than CNN-based methods on all datasets. This may derive from transformer's advantage in capturing global information and long-range dependencies, enabling the model to more accurately understand the contextual information of the target in the image. For Ours-Hyb, it takes full advantage of both CNN and transformer, combining the features of both to show the best performance. This combination is able to capture various features of the target more comprehensively in different scenarios, thus improving the performance and robustness of the model.

In summary, the advantages of different backbone networks positively affect the performance of our approach, further demonstrating the flexibility and effectiveness of our approach.

4) Computational Complexity Comparison: We measure computational complexity in two terms, both in terms of network parameters and FLOPs. The results of all methods were obtained by testing the available public code and they are reported in Table II. In summary, the network parameters and FLOPs of our method are in the mid-range. However, compared to ACCoNet, which has essentially the same network parameters and FLOPs, for example, #Param: 99.05M (Ours) versus 102.55M (ACCoNet) and FLOPs: 221.46M (Ours) versus 184.27M (ACCoNet), our performance improves significantly. The quantitative comparison and the computational complexity comparison lead to the conclusion that our method is effective.

5) Visual Comparison: We show in Fig. 8 some visual comparisons between WeightNet and 13 representative state-of-the-art methods on challenging ORSI scenarios. Since the dedicated ORSI-SOD has the best performance, we selected 12 state-of-the-art CNN-based ORSI-SODs and one CNN-based NSI-SOD method with outstanding relative performance for demonstration. The list of challenging situations we have listed is as follows: shaded targets, multiple small targets, irregularly shaped targets, targets in complex backgrounds, and targets at multiple scales. For all these scenarios, the proposed WeightNet clearly highlights all the salient targets; while other comparison methods are disturbed by the scenarios and fail in individual cases. To be specific, in scenes with shadows, only our method can detect complete targets, while other methods will treat partial shadows as targets. In the detection of multiple small targets, only our method can clearly detect all small targets, and each small target has clear boundary and finer details. AESINet, MJRBM, and PA-KPN can also detect all small targets, but none of the edges is clear enough. In irregularly shaped target detection,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:13:46 UTC from IEEE Xplore. Restrictions apply.

TABLE V

QUANTITATIVE ABLATION ANALYSIS OF THE INDIVIDUAL CONTRIBUTIONS OF MWFAM, MDFGM, AND MPD IN WEIGHTNET. THE BASELINE IS THE ENCODER–DECODER NETWORK WITH VGG-16 AS THE BACKBONE. THE BEST RESULTS IN EACH METRIC ARE EMPHASIZED IN **BOLD**

No.	Baseline	MWFAM	MDFGM	MPD	EORSSD Dataset				ORSSD Dataset			
					$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
1	✓				.9113	.8594	.9568	.0098	.9166	.8878	.9608	.0127
2	✓	✓			.9295	.8815	.9684	.0072	.9378	.9060	.9746	.0090
3	✓	✓	✓		.9326	.8859	.9721	.0067	.9410	.9121	.9773	.0084
4	✓	✓		✓	.9337	.8873	.9728	.0065	.9421	.9135	.9786	.0083
5	✓	✓	✓	✓	**.9384**	**.8907**	**.9756**	**.0063**	**.9481**	**.9188**	**.9817**	**.0075**

our method can capture the complete shape of the target while still achieving the detail distinction. In both complex and multiscale scenes, our method has outstanding performance. To summarize, our method shows strong adaptability and robustness in these scenarios, both in terms of correct target localization and target contour details.

D. Ablation Studies

In this section, we conduct comprehensive ablation experiments to evaluate the effectiveness of our WeightNet essential components on the EORSSD and ORSSD datasets. On a more specific note, we analyze: 1) the individual contributions of MWFAM, MDFGM, and MPD in WeightNet; 2) the efficiency of weighted multiscale features in MWFAM; 3) the efficiency of weighted high-dimensional features in MDFGM; and 4) the complementarity of the synthesized loss functions. For each variant experiment, we rigorously retrain it using the same parameter settings and dataset as in Section IV-A.

1) Individual Contributions of MWFAM, MDFGM, and MPD in WeightNet: As shown in Table V, in order to evaluate the individual contributions of the proposed key components (MWFAM, MDFGM, and MPD) to WeightNet, we provide four variants: 1) the encoder–decoder network (i.e., baseline); 2) the baseline network with only MWFAM (i.e., baseline + MWFAM); 3) a combination of the baseline network with MWFAM and MDFGM (i.e., baseline + MWFAM + MDFGM); and 4) a combination of the baseline network with MWFAM and MPD (i.e., baseline + MWFAM + MPD). Furthermore, the complete WeightNet is "baseline + MWFAM + MDFGM + MPD." We report the quantitative results for each variant in Table V.

We observed, based on the quantitative performance in Table V, that each module of WeightNet contributes positively to the improvement of the performance of the baseline network. Taking the EORSSD dataset as an example, the full WeightNet improves the original baseline by 2.97%, 3.64%, 1.94%, and 0.0035% on S_α, $F_\beta^{\max}$, $E_\xi^{\max}$, and $\mathcal{M}$, respectively. MWFAM, the core component of WeightNet, improves the baseline's $F_\beta^{\max}$ and $E_\xi^{\max}$ by 2.57% and 1.21%. MDFGM improves $F_\beta^{\max}$ and $E_\xi^{\max}$ of the baseline + MWFAM by 0.49% and 0.38%, respectively. With the cooperation of MWFAM and MPD, the baseline's $F_\beta^{\max}$ and $E_\xi^{\max}$ were improved by 3.24% and 1.67%, respectively. The trend on the ORSSD dataset was the same as that on the EORSSD dataset. Thus, the above analysis verifies that each module in WeightNet is available for ORSI-SOD.

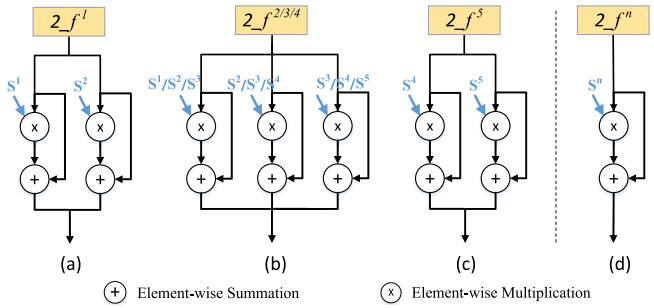


Fig. 9. Structures of the two types of MWFAM variants. (a)–(c) Structure of the first variant. (d) Structure of the second variant.

2) Efficiency of Weighted Multiscale Features in MWFAM: For the purpose of justifying the multiscale weighting of features in MWFAM, we utilize two strategies to change the number of multiscale weighted branches (the blue dashed part in Fig. 5).

1) Replace multiscale weighted branches with adjacent scale weighted branches (i.e., w/AB). The specific implementation strategy is shown in Fig. 9, with two adjacent weighted branches for the first and the fifth layer features [Fig. 9(a) and (c)], and three weighted branches for the second, third, and fourth layer features [Fig. 9(b)].
2) Replace the multiscale weighted branches with weighted branches of the same scale (i.e.,w/CB), as shown in Fig. 9(d). The results of the ablation experiments are reported in Table VI.

For the case of w/AB, the performance of $F_\beta^{\max}$ and $E_\xi^{\max}$ decreases by 0.84% and 0.41% on the EORSSD dataset, respectively; $\mathcal{M}$ increases by 0.0015. On the ORSSD dataset, the performance of $F_\beta^{\max}$ and $E_\xi^{\max}$ decreases by 0.62% and 0.37%, respectively; $\mathcal{M}$ increases by 0.0013. In the case of w/CB, on the EORSSD dataset, $F_\beta^{\max}$ and $E_\xi^{\max}$ decrease by 0.84% and 0.41%, respectively, and $\mathcal{M}$ increases by 0.0015. On the ORSSD dataset, $F_\beta^{\max}$ and $E_\xi^{\max}$ decrease by 0.62% and 0.37%, respectively, and $\mathcal{M}$ increases by 0.0013. It is observed that the more scale information is enriched, the better the results are. It also shows that our MWFAM takes more comprehensive scale information into account.

3) Efficiency of Weighted High-Dimensional Features in MDFGM: In an attempt to verify the reasonableness of the high-dimensional features being weighted in MDFGM, we cancel the weighting operation of the high-dimensional features. The high-dimensional features are directly used as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:13:46 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
QUANTITATIVE ABLATION STUDY ON THE WEIGHTING APPROACH OF MWFAM AND MDFGM IN WEIGHTED FEATURE AGGREGATION NETWORK

Models	EORSSD Dataset			ORSSD Dataset		
	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
$w/\ AB$	.8835	.9720	.0075	.9127	.9775	.0092
$w/\ CB$	.8803	.9672	.0094	.9096	.9722	.0106
$w/o\ DW$	.8832	.9716	.0078	.9131	.9780	.0088
WeightNet	**.8907**	**.9756**	**.0063**	**.9188**	**.9817**	**.0075**

TABLE VII
QUANTITATIVE ABLATION STUDIES ON THE COMPLEMENTARITY OF THE SYNTHETIC LOSS FUNCTION (BCE + IOU + F-m). THE BEST RESULTS IN EACH METRIC ARE HIGHLIGHTED IN **BOLD**

No.	BCE	IoU	F-m	EORSSD		ORSSD	
				$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$
1	✓			.8765	.9728	.9122	.9746
2	✓	✓		.8879	.9746	.9163	.9793
3	✓		✓	.8864	.9737	.9144	.9779
4	✓	✓	✓	**.8907**	**.9756**	**.9188**	**.9817**

the guidance information, which is also the way adopted by PoolNet [27]. The results of the ablation experiments are reported in the third row of Table VI (i.e., w/oDW).

From the results, it can be seen that the network performance decreases after removing the weighting of the high-dimensional features. On the EORSSD dataset, the performance of $F_\beta^{\max}$ and $E_\xi^{\max}$ decreases by 0.84% and 0.41%, respectively; and $\mathcal{M}$ increases by 0.0015. On the ORSSD dataset, the performance of $F_\beta^{\max}$ and $E_\xi^{\max}$ decreases by 0.62% and 0.37%, respectively; and $\mathcal{M}$ increases by 0.0013. The above data fully demonstrate the feasibility of weighted guidance for high-dimensional features.

4) Complementarity of the Synthesized Loss Functions: To justify the effectiveness of our combined loss function, we provide three loss variables for network training: 1) single BCE loss; 2) BCE loss and IoU loss; and 3) BCE loss with F-m loss. We present quantitative results reported in Table VII.

WeightNet achieves tolerable performance when the training strategy is a single BCE loss. Concretely, on the EORSSD dataset, $F_\beta^{\max}$ is 0.8765 and $E_\xi^{\max}$ is 0.9728. By adding the IoU loss to the BCE loss for training, $F_\beta^{\max}$ improves by 0.0114 and $E_\xi^{\max}$ increases by 0.0018. The addition of the F-m loss to the BCE loss improves $F_\beta^{\max}$ by 0.0099 and $E_\xi^{\max}$ by 0.0009. $F_\beta^{\max}$ is 0.9122 and $E_\xi^{\max}$ is 0.9746 when trained with only BCE loss on the ORSSD dataset. With the help of IoU loss or F-m loss, $F_\beta^{\max}$ increased by 0.0041, 0.0022, and $E_\xi^{\max}$ increased by 0.0047, 0.0033, respectively. Combining these three losses together to train our WeightNet yields the best performance. $F_\beta^{\max}$ and $E_\xi^{\max}$ on the EORSSD dataset increase by 0.0142 and 0.0028, and on the ORSSD dataset increase by 0.0066 and 0.0071, respectively. The above analyses clearly validate the effectiveness of the synthesized loss function.

V. DISCUSSION

In the task of ORSI-SOD, multiscale feature fusion is a common technique aimed at integrating feature information from different scales to enhance segmentation performance. However, disparities in spatial resolution and semantic information among features from different scales may lead to scale mismatches, thereby introducing inconsistency and noise between features. Additionally, features from different scales may contain conflicting information, such as low-level features focusing on edge details while high-level features emphasize semantic information, resulting in conflicts during feature fusion and hindering accurate learning.

To address these issues, we propose a two-stage WeightNet approach for ORSI-SOD. The first-stage network generates saliency maps at different scales to provide weighted information for the second stage. The key component, MWFAM, in the second stage network indirectly performs multiscale weighting on original features, effectively avoiding noise interference caused by direct interaction and fusion of multiscale features. MWFAM weights the original features at each scale with information from five scales to capture global information of salient objects. It further enhances local features by using residual blocks with small convolution kernels, effectively capturing the global–local features of salient objects. Furthermore, to refine the details and edges of salient objects, we design MDFGM, which utilizes low-dimensional saliency maps to weight guiding features with high-dimensional semantic information, highlighting the details and contour information of salient objects while suppressing interference from cluttered backgrounds. Experimental results demonstrate that our design not only improves the accuracy of SOD but also provides richer details.

However, our method also has some limitations. In scenarios where real-time performance is highly demanded, our method may require longer inference time and may not meet real-time requirements. Additionally, due to the difficulty in obtaining large-scale annotated ORSI data, insufficient training data may limit the generalization ability of the model.

VI. CONCLUSION

In this article, we propose an effective two-stage feature weighting network (WeightNet) to model the multiscale information of ORSI data for the multiscale characteristics. The network consists of three key components: MWFAM, MDFGM, and MPD. In MWFAM, five-scale weighted features can highlight the target area more completely without adding noise. In MDFGM, high-dimensional features can provide accurate positional guidance, while low-dimensional features can provide more detailed information to accurately localize the target while sharpening the details. Meanwhile, MPD can capture more scale information, which enhances the inference ability of the decoder. In addition, we equip these components into an encoder–decoder network and propose a complete solution, WeightNet, for ORSI-SOD. Finally, a comprehensive loss function is employed in the training stage to improve the training performance. We experimented extensively on three public ORSI-SOD datasets. The results of the experiments demonstrate the superiority of this WeightNet and the effectiveness of the individual modules.

In the next work, WeightNet will be deployed for the task of compression of ROI for optical remote sensing images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:13:46 UTC from IEEE Xplore. Restrictions apply.

In addition, we will explore lightweight networks that are more suitable for real-world scenarios with a view to optimizing the network.

REFERENCES

[1] W. Wang, Q. Lai, H. Fu, J. Shen, H. Ling, and R. Yang, "Salient object detection in the deep learning era: An in-depth survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 6, pp. 3239–3259, Jun. 2022.

[2] W. Wang, J. Shen, J. Xie, M.-M. Cheng, H. Ling, and A. Borji, "Revisiting video saliency prediction in the deep learning era," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 1, pp. 220–237, Jan. 2021.

[3] A. Borji, M.-M. Cheng, H. Jiang, and J. Li, "Salient object detection: A benchmark," *IEEE Trans. Image Process.*, vol. 24, no. 12, pp. 5706–5722, Dec. 2015.

[4] A. Borji, M.-M. Cheng, Q. Hou, H. Jiang, and J. Li, "Salient object detection: A survey," *Comput. Vis. Media*, vol. 5, no. 2, pp. 117–150, Jun. 2019.

[5] Y. Gu, L. Wang, Z. Wang, Y. Liu, M. Cheng, and S. Lu, "Pyramid constrained self-attention network for fast video salient object detection," in *Proc. 34th AAAI Conf. Artif. Intell.*, vol. 34, no. 7, 2020, pp. 10869–10876.

[6] G. Li et al., "Personal fixations-based object segmentation with object localization and boundary preservation," *IEEE Trans. Image Process.*, vol. 30, pp. 1461–1475, 2021.

[7] Y. Fang, Z. Chen, W. Lin, and C.-W. Lin, "Saliency detection in the compressed domain for adaptive image retargeting," *IEEE Trans. Image Process.*, vol. 21, no. 9, pp. 3888–3901, Sep. 2012.

[8] Y. Fang, W. Lin, Z. Chen, C.-M. Tsai, and C.-W. Lin, "A video saliency detection model in compressed domain," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 24, no. 1, pp. 27–38, Jan. 2014.

[9] P. Zhang, W. Liu, D. Wang, Y. Lei, H. Wang, and H. Lu, "Non-rigid object tracking via deep multi-scale spatial–temporal discriminative saliency maps," *Pattern Recognit.*, vol. 100, Apr. 2020, Art. no. 107130.

[10] Y.-C. Chen, K.-J. Chang, Y.-H. Tsai, Y.-C. F. Wang, and W.-C. Chiu, "Guide your eyes: Learning image manipulation under saliency guidance," in *Proc. BMVC*, vol. 2, 2019, pp. 1–12.

[11] X. Wang, L. Ma, S. Kwong, and Y. Zhou, "Quaternion representation based visual saliency for stereoscopic image quality assessment," *Signal Process.*, vol. 145, pp. 202–213, Apr. 2018.

[12] J. Han, X. Yao, G. Cheng, X. Feng, and D. Xu, "P-CNN: Part-based convolutional neural networks for fine-grained visual categorization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 2, pp. 579–590, Feb. 2022.

[13] Y. Wang, R. Wang, X. Fan, T. Wang, and X. He, "Pixels, regions, and objects: Multiple enhancement for salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 10031–10040.

[14] H. Zhu et al., "I can find you! Boundary-guided separated attention network for camouflaged object detection," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2022, pp. 3608–3616.

[15] X. Tian, J. Zhang, M. Xiang, and Y. Dai, "Modeling the distributional uncertainty for salient object detection models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 19660–19670.

[16] Z. Wang, Y. Ma, and Y. Zhang, "Review of pixel-level remote sensing image fusion based on deep learning," *Inf. Fusion*, vol. 90, pp. 36–58, Feb. 2023.

[17] W. Miao, J. Geng, and W. Jiang, "Multigranularity decoupling network with pseudolabel selection for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5603813.

[18] Y. Ji, H. Zhang, Z. Zhang, and M. Liu, "CNN-based encoder–decoder networks for salient object detection: A comprehensive review and recent advances," *Inf. Sci.*, vol. 546, pp. 835–857, Feb. 2021.

[19] X. Zeng, M. Xu, Y. Hu, H. Tang, Y. Hu, and L. Nie, "Adaptive edge-aware semantic interaction network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5617416.

[20] G. Li, Z. Liu, W. Lin, and H. Ling, "Multi-content complementation network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5614513.

[21] Q. Zhang et al., "Dense attention fluid network for salient object detection in optical remote sensing images," *IEEE Trans. Image Process.*, vol. 30, pp. 1305–1317, 2021.

[22] Q. Hou, M.-M. Cheng, X. Hu, A. Borji, Z. Tu, and P. Torr, "Deeply supervised salient object detection with short connections," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3203–3212.

[23] Y. Pang, X. Zhao, L. Zhang, and H. Lu, "Multi-scale interactive network for salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9413–9422.

[24] G. Li, Z. Liu, D. Zeng, W. Lin, and H. Ling, "Adjacent context coordination network for salient object detection in optical remote sensing images," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 526–538, Jan. 2023.

[25] P. Zhang, D. Wang, H. Lu, H. Wang, and X. Ruan, "Amulet: Aggregating multi-level convolutional features for salient object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 202–211.

[26] T. Wang et al., "Detect globally, refine locally: A novel approach to saliency detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3127–3135.

[27] J.-J. Liu, Q. Hou, M.-M. Cheng, J. Feng, and J. Jiang, "A simple pooling-based design for real-time salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 3917–3926.

[28] X. Zhou et al., "Edge-guided recurrent positioning network for salient object detection in optical remote sensing images," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 539–552, Jan. 2023.

[29] D. Feng et al., "Boundary-semantic collaborative guidance network with dual-stream feedback mechanism for salient object detection in optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4706317.

[30] L. Itti, C. Koch, and E. Niebur, "A model of saliency-based visual attention for rapid scene analysis," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 20, no. 11, pp. 1254–1259, Sep. 1998.

[31] Z. Liu, R. Shi, L. Shen, Y. Xue, K. N. Ngan, and Z. Zhang, "Unsupervised salient object segmentation based on kernel density estimation and two-phase graph cut," *IEEE Trans. Multimedia*, vol. 14, no. 4, pp. 1275–1289, Aug. 2012.

[32] Z. Liu, W. Zou, and O. Le Meur, "Saliency tree: A novel saliency detection framework," *IEEE Trans. Image Process.*, vol. 23, no. 5, pp. 1937–1952, May 2014.

[33] C. Li, Y. Yuan, W. Cai, Y. Xia, and D. D. Feng, "Robust saliency detection via regularized random walks ranking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 2710–2717.

[34] Y. Yuan, C. Li, J. Kim, W. Cai, and D. D. Feng, "Reversion correction and regularized random walk ranking for saliency detection," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1311–1322, Mar. 2018.

[35] W. Zou, Z. Liu, K. Kpalma, J. Ronsin, Y. Zhao, and N. Komodakis, "Unsupervised joint salient region detection and object segmentation," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 3858–3873, Nov. 2015.

[36] J. Kim, D. Han, Y.-W. Tai, and J. Kim, "Salient region detection via high-dimensional color transform and local spatial support," *IEEE Trans. Image Process.*, vol. 25, no. 1, pp. 9–23, Jan. 2016.

[37] L. Zhou, Z. Yang, Z. Zhou, and D. Hu, "Salient region detection using diffusion process on a two-layer sparse graph," *IEEE Trans. Image Process.*, vol. 26, no. 12, pp. 5882–5894, Dec. 2017.

[38] H. Peng, B. Li, H. Ling, W. Hu, W. Xiong, and S. J. Maybank, "Salient object detection via structured matrix decomposition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 818–832, Apr. 2017.

[39] M. Huang, Z. Liu, L. Ye, X. Zhou, and Y. Wang, "Saliency detection via multi-level integration and multi-scale fusion neural networks," *Neurocomputing*, vol. 364, pp. 310–321, Oct. 2019.

[40] X. Zhao, Y. Pang, L. Zhang, H. Lu, and L. Zhang, "Suppress and balance: A simple gated network for salient object detection," in *Proc. Eur. Conf. Comput. Vis.*, Aug. 2020, pp. 35–51.

[41] K. Yan, X. Wang, J. Kim, and D. Feng, "A new aggregation of DNN sparse and dense labeling for saliency detection," *IEEE Trans. Cybern.*, vol. 51, no. 12, pp. 5907–5920, Dec. 2021.

[42] J. Zhao, J.-J. Liu, D.-P. Fan, Y. Cao, J. Yang, and M.-M. Cheng, "EGNet: Edge guidance network for salient object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8779–8788.

[43] H. Zhou, X. Xie, J.-H. Lai, Z. Chen, and L. Yang, "Interactive two-stream decoder for accurate and fast saliency detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9141–9150.

[44] Z. Chen, Q. Xu, and R. Cong, "Global context-aware progressive aggregation network for salient object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 7, pp. 10599–10606.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:13:46 UTC from IEEE Xplore. Restrictions apply.

[45] Q. Zhang, L. Zhang, W. Shi, and Y. Liu, "Airport extraction via complementary saliency analysis and saliency-oriented active contour model," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 7, pp. 1085–1089, Jul. 2018.

[46] Z. Liu, D. Zhao, Z. Shi, and Z. Jiang, "Unsupervised saliency model with color Markov chain for oil tank detection," *Remote Sens.*, vol. 11, no. 9, p. 1089, May 2019.

[47] L. Zhang, Y. Wang, and Y. Sun, "Salient target detection based on the combination of super-pixel and statistical saliency feature analysis for remote sensing images," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 2336–2340.

[48] L. Zhang, Y. Liu, and J. Zhang, "Saliency detection based on self-adaptive multiple feature fusion for remote sensing images," *Int. J. Remote Sens.*, vol. 40, no. 22, pp. 8270–8297, Nov. 2019.

[49] D. Faur, I. Gavat, and M. Datcu, "Salient remote sensing image segmentation based on rate-distortion measure," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 855–859, Oct. 2009.

[50] C. Li, R. Cong, J. Hou, S. Zhang, Y. Qian, and S. Kwong, "Nested network with two-stream pyramid for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9156–9166, Nov. 2019.

[51] X. Zhou, K. Shen, Z. Liu, C. Gong, J. Zhang, and C. Yan, "Edge-aware multiscale feature integration network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5605315.

[52] Z. Tu, C. Wang, C. Li, M. Fan, H. Zhao, and B. Luo, "ORSI salient object detection via multiscale joint region and boundary model," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607913.

[53] Q. Zheng, L. Zheng, Y. Bai, H. Liu, J. Deng, and Y. Li, "Boundary-aware network with two-stage partial decoders for salient object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605713.

[54] D. Song, Y. Dong, and X. Li, "Adjacent complementary network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5403111.

[55] B. Liang and H. Luo, "MEANet: An effective and lightweight solution for salient object detection in optical remote sensing images," *Expert Syst. Appl.*, vol. 238, Mar. 2024, Art. no. 121778.

[56] G. Li, Z. Liu, X. Zhang, and W. Lin, "Lightweight salient object detection in optical remote-sensing images via semantic matching and edge alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5601111.

[57] G. Li, Z. Liu, Z. Bai, W. Lin, and H. Ling, "Lightweight salient object detection in optical remote sensing images via feature correlation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5617712.

[58] Q. Wang, Y. Liu, Z. Xiong, and Y. Yuan, "Hybrid feature aligned network for salient object detection in optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5624915.

[59] L. Bao et al., "Aggregating transformers and CNNs for salient object detection in optical remote sensing images," *Neurocomputing*, vol. 553, Oct. 2023, Art. no. 126560.

[60] Z. Bai, G. Li, and Z. Liu, "Global–local–global context-aware network for salient object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 198, pp. 184–196, Apr. 2023.

[61] Y. Zhang, J. Guo, H. Yue, X. Yin, and S. Zheng, "Transformer guidance dual-stream network for salient object detection in optical remote sensing images," *Neural Comput. Appl.*, vol. 35, no. 24, pp. 17733–17747, Aug. 2023.

[62] J. Li, Z. Pan, Q. Liu, Y. Cui, and Y. Sun, "Complementarity-aware attention network for salient object detection," *IEEE Trans. Cybern.*, vol. 52, no. 2, pp. 873–886, Feb. 2022.

[63] Z. Huang, H. Chen, B. Liu, and Z. Wang, "Semantic-guided attention refinement network for salient object detection in optical remote sensing images," *Remote Sens.*, vol. 13, no. 11, p. 2163, May 2021.

[64] S. Chen, B. Wang, X. Tan, and X. Hu, "Embedding attention and residual network for accurate salient object detection," *IEEE Trans. Cybern.*, vol. 50, no. 5, pp. 2050–2062, May 2020.

[65] T. Zhao and X. Wu, "Pyramid feature attention network for saliency detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3085–3094.

[66] Y. Gu, H. Xu, Y. Quan, W. Chen, and J. Zheng, "ORSI salient object detection via bidimensional attention and full-stage semantic guidance," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5603213.

[67] J. Zheng, Y. Quan, H. Zheng, Y. Wang, and X. Pan, "ORSI salient object detection via cross-scale interaction and enlarged receptive field," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

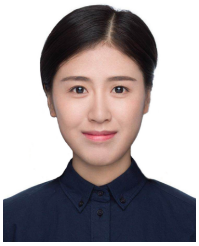
[68] Z. Deng et al., "R^3Net: Recurrent residual refinement network for saliency detection," in *Proc. 27th Int. Joint Conf. Artif. Intell.*, Palo Alto, CA, USA, Jul. 2018, pp. 684–690.

[69] B. Xu, H. Liang, R. Liang, and P. Chen, "Locate globally, segment locally: A progressive architecture with knowledge review network for salient object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 4, pp. 3004–3012.

[70] H. Luo and B. Liang, "Semantic-edge interactive network for salient object detection in optical remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 6980–6994, 2023.

[71] Y. Liu, D. Zhang, N. Liu, S. Xu, and J. Han, "Disentangled capsule routing for fast part-object relational saliency," *IEEE Trans. Image Process.*, vol. 31, pp. 6719–6732, 2022.

[72] C. Fang, H. Tian, D. Zhang, Q. Zhang, J. Han, and J. Han, "Densely nested top-down flows for salient object detection," *Sci. China Inf. Sci.*, vol. 65, no. 8, Aug. 2022, Art. no. 182103.



Lamei Di is currently pursuing the Ph.D.degree with the School of Software Engineering, Xi'an Jiaotong University, Xi'an, China.

Her research interests include salient object detection and compression for optical remote sensing images.



Bin Zhang (Member, IEEE) received the Ph.D. degree in pattern recognition and intelligence systems from Xi'an Jiaotong University, Xi'an, China, in 2013.

He is a Full Professor with the School of Software, Xi'an Jiaotong University. His research interests include image processing, computer vision, and VLSI design.



Yiming Wang received the master's degree from the School of Software Engineering, Xi'an Jiaotong University, Xi'an, China, in 2019, where he is currently pursuing the Ph.D. degree.

His research interests include EEG-based emotion analysis and domain generalization.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:13:46 UTC from IEEE Xplore. Restrictions apply.